

HAT-P-27b

R-0076

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_250419 · created 2026-07-15T19:50:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460784.9377 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.061 +/- 0.04
Transit depth (Rp/Rs)^2	0.37 +/- 0.48 [%]
Orbital Inclination (inc)	89.34 +/- 2.85
Airmass coefficient 1 (a1)	0.993 +/- 0.011
Airmass coefficient 2 (a2)	0.007 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#2 - [287, 281]
Optimal Aperture	3.14
Optimal Annulus	6.49
Transit Duration (day)	0.052 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.061 $\pm$ 0.04 vs 0.119 (deviation 1.45 σ)
Duration fit vs published	0.052 d vs 0.07 d (deviation 1.38 σ)
Mid-time O-C	21.8 min
Depth SNR	1.5
Residual scatter	1.15 %

Light curve

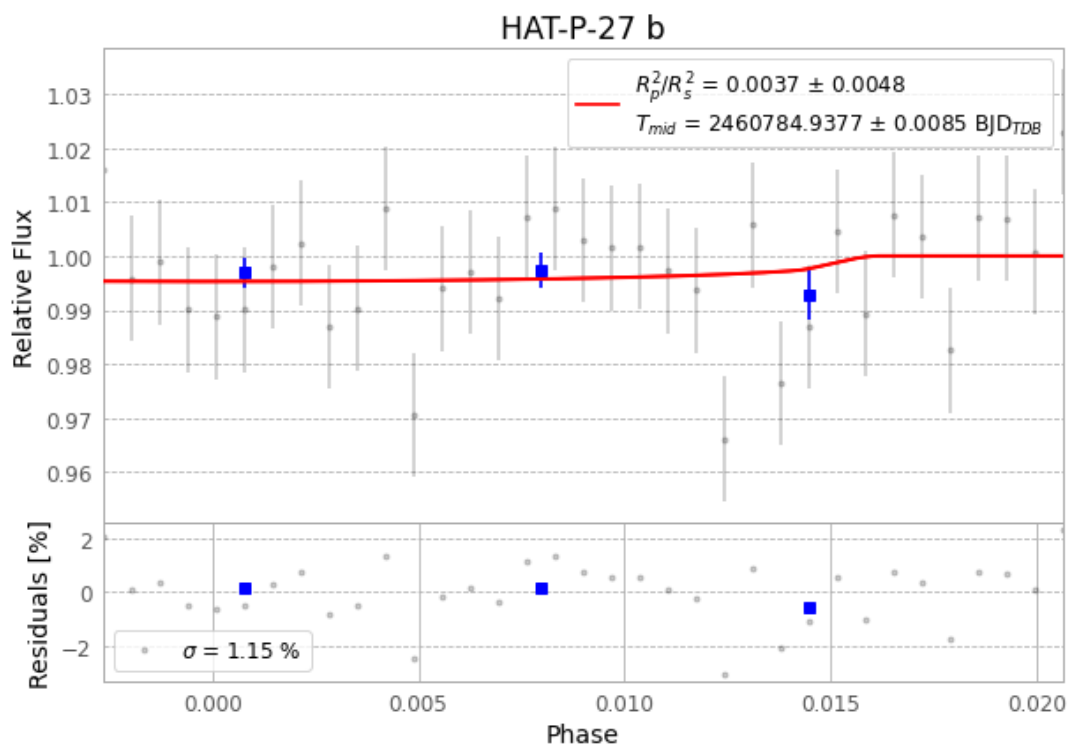


Figure 1 — FinalLightCurve_HAT-P-27 b_2025-04-19.png (SHA-256 ed59e2aa4bd23787...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 283], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4b7d4d2930422f27...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

35 FITS frames (23 MB), HATP-27250419100916.FITS → HATP-27250419115115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-250419.log (fd7f0cc12e15...), FinalLightCurve_HAT-P-27 b_2025-04-19.png (ed59e2aa4bd2...), FinalLightCurve_HAT-P-27 b_2025-04-19.csv (597711a1a6ce...)

Compute resources

Wall time 175.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 19:50Z.